

SAFETY DATA SHEET**1. CHEMICAL PRODUCT AND COMPANY IDENTIFICATION**

Product Name	PENGUIN CEMENT 1081LM
Manufacturer Name	Sunstar Chemical (Thailand) Co., Ltd.
Address	624 Moo 4 Bangpoo Industrial Estate Soi 7A Sukhumvit Rd., Praksa, Muang Samutprakarn, Samutprakarn 10280 Thailand
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2. HAZARDS IDENTIFICATION**PHYSICAL HAZARDS**

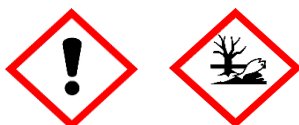
Not classified

HEALTH HAZARDS

Acute toxicity (Oral)	Not classified
Acute toxicity (Dermal)	Category 4
Acute toxicity (Inhalation: Gases)	Not applicable
Acute toxicity (Inhalation: Vapours)	Classification not possible
Acute toxicity (Inhalation: Dusts and mists)	Classification not possible
Skin corrosion/irritation	Category 2
Serious eye damage/eye irritation	Category 2A
Respiratory sensitization	Classification not possible
Skin sensitization	Category 1
Germ cell mutagenicity	Not classified
Carcinogenicity	Not classified
Reproductive toxicity	Not classified
Specific target organ toxicity - Single exposure	Not classified
Specific target organ toxicity - Repeated exposure	Not classified
Aspiration hazard	Classification not possible

ENVIRONMENTAL HAZARDS

Hazardous to the aquatic environment (Acute)	Category 1
Hazardous to the aquatic environment (Long-term)	Category 1
Hazardous to the ozone layer	Classification not possible

ALLOCATION OF LABEL ELEMENTS**Hazard symbol****Signal word**

Warning

Hazard statement

H312 Harmful in contact with skin.
H315 Causes skin irritation.
H317 May cause an allergic skin reaction.
H319 Causes serious eye irritation.
H400 Very toxic to aquatic life.
H410 Very toxic to aquatic life with long lasting effects.

Precautionary statements**Prevention**

P261 Avoid breathing dust/fume/gas/mist/vapours/spray.
P264 Wash thoroughly after handling.
P272 Contaminated work clothing should not be allowed out of the workplace.
P273 Avoid release to the environment.
P280 Wear protective gloves/protective clothing/eye protection/face protection.

Response

- P312 Call a POISON CENTER or doctor/physician if you feel unwell.
P321 Specific treatment (see label).
P391 Collect spillage.
P302+P352 IF ON SKIN: Wash with plenty of soap and water.
P332+P313 If skin irritation occurs: Get medical advice/attention.
P333+P313 If skin irritation or rash occurs: Get medical advice/attention.
P337+P313 If eye irritation persists: Get medical advice/attention.
P362+P364 Take off contaminated clothing and wash it before reuse.
P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

Disposal

- P501 Dispose of contents/container to..... (in accordance with local/regional/national/international regulation).

3.COMPOSITION/INFORMATION ON INGREDIENTS

Substance/Preparation	Preparation	
General Name	Epoxy Type Adhesives	
Components Contributing to the Hazard		
Components	Cas No	%WT
Epoxy resin	25068-38-6	30-40
Calcium oxide	1305-78-8	<5
Glycidyl ether of C ₁₂₋₁₄ alcohols	68609-97-2	<5

4.FIRST AND MEASURES**Inhalation**

Remove to fresh air and keep at rest in a position comfortable for breathing.

Skin Contact

Wash with plenty of soap and water.

If skin irritation or rash occurs: Get medical advice/attention.

Eye Contact

Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

If eye irritation persists: Get medical advice/attention.

Ingestion

Get medical attention/advice.

5.FIRE-FIGHTING**Extinguishing Media**

Carbon dioxide, foam, dry sand

Special Hazards

No data

Special Methods

Evacuate non essential personnel.

Move container from fire areas if it can be done without risk.

Keep personnel removed from and upwind of fire.

Protection of Firefighters

Wear self-contained breathing apparatus.

6.ACCIDENTAL RELEASE MEASURES**Personal Precautions**

Wear protective equipment and avoid contact with skin and eye.

Environmental Precautions

Avoid release to the environment.

Methods for Cleaning up

Absorb with liquid-binding materials (sand, sawdust etc.).

Scoop material into containers for disposal.

7.HANDLING AND STORAGE

Handling

Technical Measures

- Avoid breathing dust/fume/gas/mist/vapours/spray.
- Wash thoroughly after handling.
- Contaminated work clothing should not be allowed out of the workplace.
- Keep containers closed when not in use.

Precautions

- Use in the well-ventilated areas.

Storage

Suitable Storage Conditions

- Keep container tightly closed.
- Store in a cool (<10°C) and well-ventilated place.
- Protect material from direct sunlight.

8.EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

Components	Cas No	NIOSH REL (TWA)
Epoxy resin	25068-38-6	Not established
Calcium oxide	1305-78-8	2 mg/m ³
Glycidyl ether of C ₁₂₋₁₄ alcohols	68609-97-2	Not established

Personal Protective Equipment

Hand Protection

- Protective gloves

Eye Protection

- Safety glasses

Skin and Body Protection

- Avoid contact with skin.

Hygiene Measures

- Wash hands before taking a break and at the end of work.

9.PHYSICAL AND CHEMICAL PROPERTIES

Physical State

Form	Paste
Color	Dark Brown
Odor	Slightly odor
pH	Not established
Melting point/freezing point	Not established
Initial boiling point and boiling range	Not established
Flash point	>150°C
Evaporation rate	Not established
Flammability	Not established
Upper/lower flammability or explosive limits	Not established
Vapour pressure	Not established
Relative density	About 1.6
Solubility	Soluble in organic solvents
Partition coefficient : n-octanol/water	Not established
Auto-ignition temperature	Not established
Viscosity	Not established

10.STABILITY AND REACTIVITY

Stability

- Stable under the condition of normal usage.

Possible Hazardous Reactions

- No data

Conditions to Avoid

High Heat

Hazardous Decomposition Products

Chlorine gas

11.TOXICOLOGICAL INFORMATION**Acute toxicity**

Epoxy resin :	LD50 (Oral) :	Rat	11,400 mg/kg	(CERI Hazard Data 2001-36 (2002))
Calcium oxide :	LD50 (Oral) :	Rat	5,000 mg/kg and 5,916 mg/kg	(Risk Assessment Report (Food Additives) (Food Safety Commission of Japan, 2013))
Glycidyl ether of C ₁₂₋₁₄ alcohols :	LD50 (Oral) :	Rat	17,100 mg/kg	(RTECS (1999))

Skin corrosion/irritation

Epoxy resin :	Category 2
Calcium oxide :	Category 2
Glycidyl ether of C ₁₂₋₁₄ alcohols :	Category 2

Serious eye damage/eye irritation

Epoxy resin :	Category 2B
Calcium oxide :	Category 1
Glycidyl ether of C ₁₂₋₁₄ alcohols :	Category 2B

Skin sensitization

Epoxy resin :	Category 1
Glycidyl ether of C ₁₂₋₁₄ alcohols :	Category 1

Germ cell mutagenicity

Epoxy resin :	Not classified
Glycidyl ether of C ₁₂₋₁₄ alcohols :	Not classified

Reproductive toxicity

Epoxy resin :	Not classified
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Specific target organ toxicity - Single exposure

Calcium oxide :	Category 1 (respiratory organs)
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Specific target organ toxicity - Repeated exposure

Calcium oxide :	Category 1 (respiratory organs)
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12.ECOLOGICAL INFORMATION**Hazardous to the aquatic environment (Acute)**

Epoxy resin :	EC50 (48H) :	Crustacea (Daphnia magna)	1.7 mg/L	(CERI/NITE Hazard Assessment Report (preliminary version) (2006) and others)
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13.DISPORSAL CONSIDERATIONS**Waste from Residues**

Follow the all regulations in your country.

Contaminated Packaging

Follow the all regulations in your country.

14.TRANSPORT INFORMATION

UN Class 9

UN No 3077

Packing group III

Specific Precautionary Transport Measures and Conditions

Follow the all regulations in your country.

15.REGULATORY INFORMATION

Follow the all regulations in your country.

16.OTHER INFORMATION

The information herein is given in good faith, but no warranty, express or implied, is made.

The information presented herein is believed to be accurate but is not warranted. Recipients are advised to confirm in advance that the information is current, applicable and suitable to there circumstances.